

Pattern Recognition / *The Discrimination Layer*

A visual systems framework for deciding what information deserves acquisition, comparison, enrichment, and influence before AI generates.

Here, discrimination means technical differentiation among information and possible actions—not social classification.

WORKING PROPOSITION · CONCEPTUAL SYNTHESIS

Make the judgment before generation visible.

Some evidence-sensitive AI workflows may benefit from an explicit, inspectable responsibility for deciding what context can influence generation. Whether that responsibility improves outcomes enough to justify its cost is an empirical question.

START HERE · ABOUT 5 MINUTES

Understand the idea

Problem, thesis, map, one example, and what remains unproven.

COMPLETE PATH · ABOUT 25 MINUTES

Inspect the system

Eleven responsibilities, relationships, failures, cases, and research horizon.

STATUS Conceptual synthesis and research agenda. Not empirical validation, peer review, deployment guidance, or a claim of scientific novelty.

SOURCE BOUNDARY Reconciled with the owner-designated live v13 site. The original v13 diagram is preserved byte-for-byte and hash-verified; the archived rendered-DOM snapshot is a reference capture, while the original standalone HTML remains unavailable.

ACCESSIBILITY This is a visual/print companion. The semantic HTML reader is the accessible canonical reading surface.

The visible answer is often where hidden decisions surface.

A model can write a polished answer from a poor evidence environment. The failure appears at the end, but the consequential choices often happened before generation.

CONCRETE PREVIEW

Nine positive articles can still trace to one launch announcement.

Repeated mentions remain separate observations, but they do not establish distinct-origin support under this packet's relation rule.

ORIGIN-ACCOUNTING RECEIPT · ORIGIN-EX-01 · VERSION 0.2

Nine observations can still represent one origin.

Fictional bundle; no live data. This static receipt records relationships and a human disposition. It does not discover provenance, establish truth, or depict a required workflow.

CLAIM UNDER REVIEW

“The tool is broadly validated.”

INSUFFICIENT · nine mentions do not establish nine distinct origins.

OBSERVATIONS	KNOWN CLUSTER	SUPPORT ORIGINS	SEPARATE ROOTS
09	01	00	02
under review	for those records	under stated relation rule	contrast only

UNKNOWN stays unknown. Do not move an unresolved origin relation into the dependent or separate-root total.

Observation ledger · nine unordered records

Rows preserve observations; their order is not a workflow or confidence ranking.

RECORD	RECORD KIND	ORIGIN RELATION	WHAT THIS COUNTS AS
O01	Report observation	Origin A · DEPENDENT	Repeats the launch announcement; not separately rooted support under this relation rule.

RECORD	RECORD KIND	ORIGIN RELATION	WHAT THIS COUNTS AS
O02	Report observation	Origin A · DEPENDENT	Repeats the launch announcement; not separately rooted support under this relation rule.
O03	Report observation	Origin A · DEPENDENT	Repeats the launch announcement; not separately rooted support under this relation rule.
O04	Report observation	Origin A · DEPENDENT	Repeats the launch announcement; not separately rooted support under this relation rule.
O05	Report observation	Origin A · DEPENDENT	Repeats the launch announcement; not separately rooted support under this relation rule.
O06	Report observation	Origin A · DEPENDENT	Repeats the launch announcement; not separately rooted support under this relation rule.
O07	Report observation	Origin A · DEPENDENT	Repeats the launch announcement; not separately rooted support under this relation rule.
O08	Report observation	Origin A · DEPENDENT	Repeats the launch announcement; not separately rooted support under this relation rule.
O09	Report observation	Origin A · DEPENDENT	Repeats the launch announcement; not separately rooted support under this relation rule.

ORIGIN RELATION KEY	SEPARATE ROOTS FOR CONTRAST
DEPENDENT · traceable to an existing artifact; preserve the observation, but do not count it as separately rooted support. INDEPENDENT-AS-STIPULATED · a separate root declared by this illustration or benchmark, not provenance discovery. UNKNOWN · unresolved; preserve it and do not guess either way.	B1 · separate root · illustrative C1 · separate root · illustrative Claim support is not assessed. These roots are not counted as support for this packet's claim.

HUMAN DISPOSITION

HOLD · VERIFY ANOTHER ORIGIN RELATION

Inspect the originating announcement, then look for a separately authored benchmark and document its origin relation before changing the claim state. No automatic admission, rejection, or truth verdict.

Illustrative only · not a reported dataset · not a provenance audit · not a system runtime. No image is required to interpret this receipt.

? What was available at all?	? Which repetitions shared one origin?
? What supported each claim?	? Was another search worth its cost?
? What entered the final context?	? Who could correct the decision?

WHAT THE LAYER SEPARATES	LAYER
Authority, support, independence, relevance, authorization, and action priority remain different judgments.	A systems responsibility, not necessarily one service, model, prompt, or box.

FROM V13 TO V14

The center of gravity moved, but the pulse remains.

V13 asked how to find the specialist comment, anomalous change, repeated unanswered question, or memory that generic workflows miss. V14 asks the harder follow-on: what gives any item the right to influence the answer?

Underweighted is a starting condition, not a conclusion.

HISTORICAL REFERENCE · V13 · NOT THE V14 SYSTEM MAP

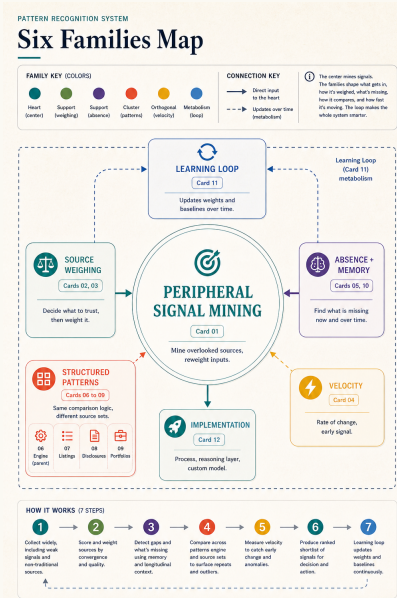
Six Families Map

The original diagram is preserved and hash-verified. It placed Peripheral Signal Mining at the center of source weighing, velocity, absence + memory, structured patterns, a learning loop, and implementation.

Historical seven-step strip

1 Collect widely. 2 Score and weight sources. 3 Detect gaps with memory and longitudinal context. 4 Compare patterns and source sets. 5 Measure velocity. 6 Produce a ranked shortlist. 7 Continuously update weights and baselines.

V14 replaces that center-and-sequence emphasis with inspectable responsibilities, typed relations, terminal states, human correction, and separately versioned outcome updates.



Keep the judgments apart.

A useful discrimination layer cannot collapse every evidential and operational question into one confidence score. These paired terms define the framework's anti-collapse rule.

Attention priority IS NOT Truth	Domain source authority IS NOT Universal trust
Claim support IS NOT Source popularity	Recurrence IS NOT Independence
Independence IS NOT Different wording, URLs, or unknown origin	Relevance IS NOT General importance
Operational authorization IS NOT Source authority or technical access	Enrichment value IS NOT Action priority or acceptance
Action priority IS NOT A factual conclusion or truth probability	Provenance IS NOT Correctness
Owner disposition IS NOT External truth	Signal candidate IS NOT A verified event or conclusion

The judgment before generation

Six families and eleven responsibilities. The flow is iterative: a gap can trigger another acquisition, a reviewer can revise the frame, and an outcome can propose—but not silently apply—a policy update.

01

Intent and authorization envelope

Define the question, consequence, permissions, baselines, and budgets before information is allowed to influence the workflow.

C01

02

Evidence spine

Acquire, identify, normalize, version, and trace artifacts without confusing capture with acceptance.

C02 · C03

03

Relationship and claim graph

Represent grouping, common origin, recurrence, comparison, atomic claims, support, contradiction, and gaps.

C04 · C05

04

Discrimination policy

Keep judgment dimensions separate, choose bounded next actions, and prepare an inspectable context packet.

C06 · C07 · C08

05

Human disposition and memory

Capture acceptance, rejection, deferral, override, and versioned retention without converting preference into fact.

C09 · C10

06

Outcome feedback

Compare expected and observed outcomes and propose revisable policy updates.

C11

RETURN Human correction and later outcomes can revise the route without rewriting the evidence.

TEXT EQUIVALENT

- A bounded question and permission envelope governs acquisition.
- Captured material receives source, artifact, version, and derivation records.
- Relationship and claim views expose origin, support, contradiction, and gaps.
- Separate assessments inform a cost-bounded next action.
- A human can correct the route; evidence and decisions remain separately versioned.
- A defined later outcome may motivate an approved policy update.

Intent and authorization envelope

Define the question, consequence, permissions, baselines, and budgets before information is allowed to influence the workflow.

C01

Decision brief and authorization envelope

WHAT IT IS A versioned declaration of the question, intended use, decision owner, stakes, expected baselines, allowed operations, sensitive-source rules, and resource limits.	
WHY IT EXISTS Relevance, meaningful absence, cost, and permissible action cannot be evaluated without a task and authority boundary.	CONSUMES owner question or decision; intended audience and use; known constraints and protected categories; privacy, retention, disclosure, and tool permissions; time, money, token, compute, and attention budgets; expected sources, perspectives, fields, or observations
PRODUCES decision_brief_version; authorization_policy_version; expected_baseline; cost_envelope; success, abstention, and escalation criteria	HOW IT INTERACTS Constrains C02 acquisition and every later transformation. Supplies the task frame for C06 relevance and action judgments. Supplies expected baselines for C04 gap detection. Defines outcomes and measurement windows used by C11.
WHAT CAN GO WRONG An underspecified question makes relevance arbitrary. Permissions are inferred from technical access. A missing baseline turns any absence into a supposed signal. Budgets omit human review or privacy exposure. The decision brief changes without invalidating downstream work.	ILLUSTRATIVE EXAMPLE · NOT A RESULT A research assistant records that it may search public sources for two hours, may not upload confidential material, must compare three named stakeholder perspectives, and must escalate unsupported high-impact claims.
EVIDENCE BOUNDARY Prior Art Supported Conceptual Synthesis. Adjacent fields: value of information, mixed initiative, decision quality, AI risk management.	WHAT REMAINS SPECULATIVE Whether one portable envelope schema can serve materially different domains. Whether an automated system can reliably detect that the brief is insufficient.
HUMAN DECISIONS approve scope; grant or deny operations; set stakes and budgets; revise or cancel the brief	COST BOUNDARIES planning time; authorized search spend; latency; tokens; review attention; privacy exposure
PROVENANCE BOUNDARIES Every downstream record references the exact brief and policy version.	OPEN QUESTIONS What is the minimum usable brief? Which changes require re-running earlier stages?

Evidence spine

Acquire, identify, normalize, version, and trace artifacts without confusing capture with acceptance.

C02

Acquisition controller

WHAT IT IS A policy-governed mechanism that proposes, authorizes, records, and stops retrieval or collection actions.	
WHY IT EXISTS Useful evidence may be absent from current context, but more acquisition has cost and risk and must not silently become belief.	CONSUMES decision brief and authorization policy; candidate sources, queries, tools, and expected information gain; current evidence gaps and uncertainty; remaining cost envelope
PRODUCES acquisition proposal; authorization decision; capture receipt or failure receipt; raw immutable artifact reference; updated remaining budget	HOW IT INTERACTS Receives gaps and enrichment proposals from C04, C05, and C07. Hands captured artifacts to C03. Reports cost and failure to C07 and C11.
WHAT CAN GO WRONG Search scope expands without authorization. Similarity or novelty is mistaken for value. Paid, private, or sensitive retrieval occurs without explicit permission. A failed capture is represented as negative evidence. The system continues searching because no stopping rule is encoded.	ILLUSTRATIVE EXAMPLE · NOT A RESULT The system proposes obtaining one primary standard because the current claim cites only commentary; it records expected benefit, estimated cost, permission, retrieval result, and why it stopped.
EVIDENCE BOUNDARY Prior Art Supported Design Hypothesis. Adjacent fields: information foraging, relevance feedback, active learning, value of information.	WHAT REMAINS SPECULATIVE Accurate expected-value estimates in open-world research. General stopping policies across domains.
HUMAN DECISIONS authorize sensitive or costly acquisition; approve expanded scope; stop or redirect search	COST BOUNDARIES money; time; latency; tokens; compute; privacy; review burden
PROVENANCE BOUNDARIES The proposal, permission, query/tool version, response, capture time, and failure state remain linked.	OPEN QUESTIONS How should unknown costs be represented? When is human escalation worth more than another retrieval?

Source, artifact, normalization, and provenance spine

WHAT IT IS A set of stable identities and append-only derivation records for sources, artifacts, captures, versions, transformations, agents, and timestamps.	
WHY IT EXISTS Comparison and audit fail when a source is confused with an artifact, normalization erases differences, or summaries become detached from origin.	CONSUMES raw capture and receipt; source observations; artifact bytes or stable reference; transformation specification; agent and tool versions
PRODUCES source identity record; artifact identity and version; content hash where available; normalized representation; provenance and derivation edges; confidence or ambiguity in identity resolution	HOW IT INTERACTS Supplies identity and derivation to C04 and C05. Travels with C08 context packets and C10 memory records. Allows C09 corrections to point to exact evidence rather than prose alone.
WHAT CAN GO WRONG Two reports are merged because names look alike. One artifact is duplicated and counted twice. Normalization removes a meaningful qualifier or time reference. A summary or embedding loses its origin binding. Mutable pages change without a capture or version record. Lineage is mistaken for correctness.	ILLUSTRATIVE EXAMPLE · NOT A RESULT A captured web page, its issuing organization, retrieval time, hash, extracted text, parser version, and paragraph offsets are separate linked records.
EVIDENCE BOUNDARY Prior Art Supported Conceptual Synthesis. Adjacent fields: W3C PROV, data lineage, evidence synthesis.	WHAT REMAINS SPECULATIVE Reliable identity resolution across aliases and mutable web artifacts. Complete provenance across opaque provider transformations.
HUMAN DECISIONS resolve ambiguous identity; approve a loss-bearing transformation; mark a source or artifact merge as uncertain	COST BOUNDARIES storage; hashing and extraction compute; manual identity resolution; retention obligations
PROVENANCE BOUNDARIES Raw captures are immutable audit evidence; normalized and derived representations are new versioned entities, not replacements.	OPEN QUESTIONS Which transformations require byte-level retention? How should inaccessible or legally restricted originals be cited and reviewed?

Relationship and claim graph

Represent grouping, common origin, recurrence, comparison, atomic claims, support, contradiction, and gaps.

C04

Relationship, recurrence, common-origin, and gap graph

WHAT IT IS

A typed graph linking sources, artifacts, events, groups, observations, time points, copies, derivations, recurrences, comparisons, and expected-but-missing perspectives.

WHY IT EXISTS

Repeated claims can share one origin, cohorts may be analytical roles, and absence or velocity is meaningful only under explicit baselines and time observations.

CONSUMES

identified artifacts and sources; provenance edges; timestamps; decision-specific grouping rules; expected baseline; observed similarities and citations

PRODUCES

typed relationship edges; candidate common-origin clusters; recurrence counts with dependence status; comparison sets; cohort role assignments; gap records; velocity observations where longitudinal data exists; signal candidates labeled as derived outputs

HOW IT INTERACTS

Uses C03 identities and provenance. Informs C05 evidence independence and contradiction analysis. Informs C06 attention and enrichment judgments. Can request C02 acquisition when a baseline gap is consequential.

WHAT CAN GO WRONG

Syndicated copies are counted as distinct-origin support. Unknown origin is silently classified as separately rooted. Topical similarity is treated as shared causal origin. A group becomes an essentialized or permanent identity. One timestamp is used to claim velocity. Unspecified expectations manufacture a meaningful absence. A signal candidate is stored or displayed as a verified event.

ILLUSTRATIVE EXAMPLE · NOT A RESULT

Nine articles repeating a vendor announcement are linked to the same press release; recurrence remains nine observations while independent-origin count remains one known origin plus any explicitly unknown origins.

EVIDENCE BOUNDARY

Prior Art Supported Design Hypothesis. Adjacent fields: evidence synthesis, provenance, coordinated-amplification research, sensemaking.

WHAT REMAINS SPECULATIVE

Automated common-origin inference at acceptable precision. Cross-domain definitions of meaningful recurrence, velocity, and gaps.

HUMAN DECISIONS

confirm or contest grouping; set expected baselines; review suspected coordination; promote or dismiss a signal candidate

COST BOUNDARIES

graph construction; entity resolution; temporal observation window; manual common-origin review

PROVENANCE BOUNDARIES

Every inferred edge records method, inputs, time, confidence, and reviewer disposition; observed and inferred edges remain distinct.

OPEN QUESTIONS

How should partial dependence be represented? What evidence is sufficient to label coordination rather than ordinary reuse?

Claim, evidence, comparison, and contradiction graph

WHAT IT IS A claim-level representation connecting atomic propositions to exact evidence spans, support states, qualifications, contradictions, alternatives, and unresolved questions.	
WHY IT EXISTS A credible source can make an unsupported claim, one artifact can contain claims with different support, and document-level ranking cannot show what a conclusion rests on.	CONSUMES artifacts and exact spans; atomic claim candidates; source and artifact identity; common-origin relationships; domain-specific evidence standards; comparison frames
PRODUCES claim records; support, contradiction, qualification, and insufficiency edges; evidence rationales; unresolved-claim list; alternative explanations; comparison matrix	HOW IT INTERACTS Uses C03 provenance and C04 independence relationships. Supplies C06 claim support and uncertainty without deciding action priority. Supplies cited material and unresolved states to C08.
WHAT CAN GO WRONG Claims remain too broad to test. Citation presence is treated as support. A source's domain authority is transferred to unrelated claims. Evidence entailment is inferred from lexical overlap. Contradictory time frames or definitions are compared as equivalents. Open-world unknowns are forced into supported or refuted.	ILLUSTRATIVE EXAMPLE · NOT A RESULT A report sentence is split into a measured result and a causal explanation; the first links to a table, while the second remains insufficient because the study design does not identify causality.
EVIDENCE BOUNDARY Prior Art Supported Conceptual Synthesis. Adjacent fields: FEVER, SciFact, FActScore, structured analytic techniques, evidence synthesis.	WHAT REMAINS SPECULATIVE Reliable domain-general atomic claim extraction. Evidence standards that transfer without flattening domain expertise.
HUMAN DECISIONS approve material claim decomposition; adjudicate contested support; choose comparison dimensions; retain unresolved states	COST BOUNDARIES claim extraction and review; span-level citation work; domain expert time
PROVENANCE BOUNDARIES Each claim version and evidence edge points to exact artifact versions and spans; generated rationales are labeled derived.	OPEN QUESTIONS What claim granularity maximizes correction without overwhelming reviewers? How should evidence standards be represented by domain?

Discrimination policy

Keep judgment dimensions separate, choose bounded next actions, and prepare an inspectable context packet.

C06

Multidimensional assessment

WHAT IT IS

An inspectable set of separate, task-scoped judgments for attention priority, domain source authority, claim support, independence, relevance, enrichment value, action priority, and owner disposition. Uncertainty and possible consequence are explicit qualifying attributes, not interchangeable scoring dimensions.

WHY IT EXISTS

Collapsing unlike judgments into one relevance or trust score hides the reasons an item can influence a decision and makes correction difficult.

CONSUMES

decision brief; source and artifact records; relationship graph; claim-evidence graph; current uncertainty; potential consequences

PRODUCES

typed assessment records; reason and evidence for each dimension; unknown or contested states; candidate action considerations; review queue

HOW IT INTERACTS

Receives task framing from C01 and evidence structure from C03-C05. Supplies assessments to C07 without making the routing decision itself. Exposes individual dimensions to C09 for correction.

WHAT CAN GO WRONG

A universal trust score launders one dimension into another. Owner interest becomes endorsement. First-party evidence is rejected categorically or trusted universally. Model confidence substitutes for evidential support. Uncertain identity or origin is hidden behind a precise score. The assessed dimension has no decision-specific definition.

ILLUSTRATIVE EXAMPLE · NOT A RESULT

A regulator's filing can have high authority for what was filed, strong support for a filing date, limited support for a company's causal explanation, high task relevance, and unknown independence from a press release—all at once.

EVIDENCE BOUNDARY

Conceptual Synthesis Requiring Evaluation. Adjacent fields: source credibility, epistemic vigilance, claim verification, relevance feedback, calibration.

WHAT REMAINS SPECULATIVE

Whether human and model assessors can apply the dimensions reliably. Whether explicit separation improves outcomes enough to justify complexity.

HUMAN DECISIONS

define domain authority criteria; contest assessments; mark unknown; decide when expert review is required

COST BOUNDARIES

assessment compute; review time; expert adjudication; interface complexity

PROVENANCE BOUNDARIES

Each assessment is a versioned interpretation with assessor, inputs, rubric, time, and confidence; it is not written into raw evidence.

OPEN QUESTIONS

Which dimensions are necessary by task type? Can ordinal states outperform misleading numerical precision?

Enrichment, stopping, and action router

WHAT IT IS

A policy that compares next actions under uncertainty and resource constraints, including acquire, compare, enrich, clarify, answer, answer provisionally, hold, defer, escalate, or refuse.

WHY IT EXISTS Assessment has no operational value until it guides a bounded next step, but selecting an action must not masquerade as a factual conclusion.	CONSUMES separate C06 assessments; evidence and gap states; authorized action set; remaining budgets; estimated value and consequence; deadline and stopping criteria
PRODUCES recommended action; ranked alternatives where appropriate; reason code; expected benefit and cost range; uncertainty; stop or escalation receipt	HOW IT INTERACTS Can request C02 acquisition or additional C04-C05 comparison. Routes selected context to C08. Requests human authorization or disposition from C09. Sends prediction and cost to C11 for later comparison.
WHAT CAN GO WRONG An enrichment-value estimate becomes acceptance or dictates action priority. A brittle utility estimate creates false precision. Search stops because a convenient answer appeared. Search never stops because certainty is impossible. The router recommends an unauthorized action. High attention priority is treated as high truth probability.	ILLUSTRATIVE EXAMPLE · NOT A RESULT For a high-impact claim with one vendor-linked source and no separately rooted support established, the router recommends one primary-source search within a fifteen-minute budget; if unavailable, it routes to a provisional answer that names the evidence gap.
EVIDENCE BOUNDARY Prior Art Supported Design Hypothesis. Adjacent fields: value of information, metareasoning, resource rationality, mixed initiative.	WHAT REMAINS SPECULATIVE Reliable utility estimation for heterogeneous evidence tasks. A portable policy vocabulary across low- and high-stakes settings.
HUMAN DECISIONS approve costly or consequential action; accept provisional output; override route; change stopping threshold	COST BOUNDARIES explicit remaining budget on every route; maximum retry or search scope; human escalation capacity; privacy and disclosure limits
PROVENANCE BOUNDARIES The route records the exact assessments, policy version, alternatives, budget, and human authorization used.	OPEN QUESTIONS Which baselines best test routing value? How should asymmetric harms and unknown utility be represented?

Bounded context packet

WHAT IT IS

A versioned package of selected content plus provenance, claim links, reasons for inclusion and exclusion, unresolved states, budgets, and generation constraints.

WHY IT EXISTS A model needs a usable context, while a reviewer needs to know what influenced it, what was withheld, and why; a long unstructured prompt provides neither guarantee.	CONSUMES authorized route; selected artifact spans and claim records; provenance; assessment dimensions; exclusions and unresolved gaps; token and disclosure budget
PRODUCES context_packet_version; selection and exclusion manifest; citation map; uncertainty and abstention instructions; generator input receipt	HOW IT INTERACTS Receives the route from C07 and exact evidence from C03-C05. Is reviewed or overridden through C09. Its use and generated output are linked in C10-C11.
WHAT CAN GO WRONG Compression changes a claim or drops a qualifier. Ordering over-amplifies convenient evidence. Exclusions are invisible. Provenance is stripped to save tokens. Context is called evidence without claim-level support. Sensitive information crosses a disclosure boundary.	ILLUSTRATIVE EXAMPLE · NOT A RESULT A packet contains three exact passages, their claim links and common-origin notes, one excluded duplicate, an unresolved counterclaim, a token budget, and an instruction to state insufficiency rather than fill the gap.
EVIDENCE BOUNDARY Design Hypothesis With Adjacent Precedent. Adjacent fields: retrieval-augmented generation, context engineering, long-context behavior, provenance.	WHAT REMAINS SPECULATIVE Optimal packet structure and ordering. How much provenance can be compressed without losing audit value.
HUMAN DECISIONS approve sensitive inclusion; review material exclusions; accept bounded uncertainty; change generation constraints	COST BOUNDARIES token limit; latency; human readability; disclosure scope
PROVENANCE BOUNDARIES Every included or transformed unit retains an origin pointer; packet version and generator invocation remain linked.	OPEN QUESTIONS What packet fields measurably improve correction? How should conflicting evidence be ordered without implying a verdict?

Human disposition and memory

Capture acceptance, rejection, deferral, override, and versioned retention without converting preference into fact.

C09

Owner disposition, review, and override

WHAT IT IS An explicit human control surface for accept, reject, defer, hold, override, request enrichment, correct relationships, and revise task constraints.	
WHY IT EXISTS The proposed responsibility is meant to remain correctable, and many judgments require domain knowledge or authority the system does not have.	CONSUMES assessment and route receipts; context packet; evidence and provenance views; uncertainties, conflicts, costs, and consequences; reviewer role and authority
PRODUCES versioned disposition; reason and optional evidence; override record; correction request; updated permission or task constraint; escalation destination	HOW IT INTERACTS Can amend C01 constraints, correct C03-C06 records, approve C07 routes, and alter C08 packets. Supplies versioned decisions to C10 and review outcomes to C11.
WHAT CAN GO WRONG Rubber-stamping or automation bias. The reviewer sees a conclusion without its evidence path. Owner preference is stored as factual truth. An override has no reason, authority, or audit record. Review burden exceeds the available attention budget. Role-based access hides necessary correction context.	ILLUSTRATIVE EXAMPLE · NOT A RESULT An analyst rejects a proposed independent-corroboration label, links both reports to a shared study, records the reason, and asks the system to recompute the route without altering the original captures.
EVIDENCE BOUNDARY Prior Art Supported Design Hypothesis. Adjacent fields: mixed initiative, human-AI interaction, structured analytic techniques, AI risk management.	WHAT REMAINS SPECULATIVE Interfaces that enable meaningful rather than ceremonial review. Portable division of authority between operator, domain expert, and system owner.
HUMAN DECISIONS accept; reject; defer; hold; override; request enrichment; escalate; revoke permission	COST BOUNDARIES review queue size; time per decision; expert availability; escalation latency
PROVENANCE BOUNDARIES Human disposition is linked to reviewer role, time, object version, displayed evidence, and reason; prior dispositions remain preserved.	OPEN QUESTIONS Which decisions require mandatory review? How can interfaces reveal uncertainty without overwhelming the reviewer?

Versioned evidence, decision, and memory ledger

WHAT IT IS Append-only retention of observations, derived interpretations, decisions, context packets, generated outputs, corrections, and supersession relationships.	
WHY IT EXISTS Learning and audit require history, while mutable memory can launder old errors or erase why a decision was made.	CONSUMES raw and derived records from C01-C09; retention and access policy; supersession and correction events; generator and prompt/model versions
PRODUCES immutable event records; current materialized views; supersession links; retrieval indexes with origin binding; audit timeline; staleness or review flags	HOW IT INTERACTS Preserves every stage without forcing all stages into one storage model. Supplies prior cases to C06-C07 only under C01 authorization. Provides the decision history that C11 compares with outcomes.
WHAT CAN GO WRONG A summary overwrites raw evidence. Retrieved memory loses source or epistemic type. An old owner preference becomes a factual prior. Sensitive records persist beyond authorization. Version proliferation makes the current state unintelligible. Poisoned or echoed content gains apparent authority through repetition.	ILLUSTRATIVE EXAMPLE · NOT A RESULT A corrected claim does not delete its earlier form; the ledger marks it superseded, preserves the evidence and decision visible at each point, and blocks the stale version from default retrieval.
EVIDENCE BOUNDARY Design Hypothesis With Adjacent Precedent. Adjacent fields: provenance, agent memory, organizational learning, data lineage.	WHAT REMAINS SPECULATIVE Safe cross-task reuse of prior decisions. Scalable provenance preservation through memory consolidation.
HUMAN DECISIONS set retention; approve cross-task reuse; correct or supersede; quarantine suspected poisoning; restore a prior view	COST BOUNDARIES storage; indexing; retention compliance; review of stale or poisoned memory
PROVENANCE BOUNDARIES Observed evidence, generated interpretation, owner disposition, and outcome are different record types; consolidation creates a derived record and never deletes origin links.	OPEN QUESTIONS What may safely transfer between tasks? How should conflicting owner dispositions be scoped and retrieved?

Outcome feedback

Compare expected and observed outcomes and propose revisable policy updates.

C11

Outcome feedback and revisable policy update

WHAT IT IS

A controlled loop that compares recorded predictions, decisions, costs, and expected outcomes with defined later observations, then proposes bounded updates.

WHY IT EXISTS

A framework cannot claim to learn without a defined outcome, measurement window, attribution boundary, and preserved prior decision state.

CONSUMES

decision and route receipts; predicted outcome or decision objective; actual cost; time-stamped outcome observation; confounders and missingness; current policy and rubric versions

PRODUCES

expected-versus-observed comparison; calibration and error records; candidate policy or rubric update; do-not-update decision; new evidence gap or research question

HOW IT INTERACTS

Reads histories from C10. Can propose but not silently apply changes to C01, C06, or C07. Routes proposed updates to C09 for human disposition.

WHAT CAN GO WRONG

No outcome was defined before the decision. A proxy is treated as the desired consequence. Hindsight creates a causal story from correlation. Selective outcome capture rewards visible successes. A single operator's preference becomes universal policy. The system updates automatically from adversarial or contaminated feedback.

ILLUSTRATIVE EXAMPLE · NOT A RESULT

The system records that an extra search was expected to resolve a source-dependence question within twenty minutes; later it records whether it did, the actual cost, and proposes a stopping-rule adjustment for owner review.

EVIDENCE BOUNDARY

Design And Empirical Hypothesis. Adjacent fields: calibration, organizational learning, decision quality, human-in-the-loop systems.

WHAT REMAINS SPECULATIVE

Causal attribution of decision outcomes in field settings. Whether outcome feedback improves routing without reinforcing local bias.

HUMAN DECISIONS

validate outcome definition; approve or reject policy update; flag confounders; scope learning to a domain or operator

COST BOUNDARIES

measurement effort; delay to outcome; expert adjudication; monitoring and retention

PROVENANCE BOUNDARIES

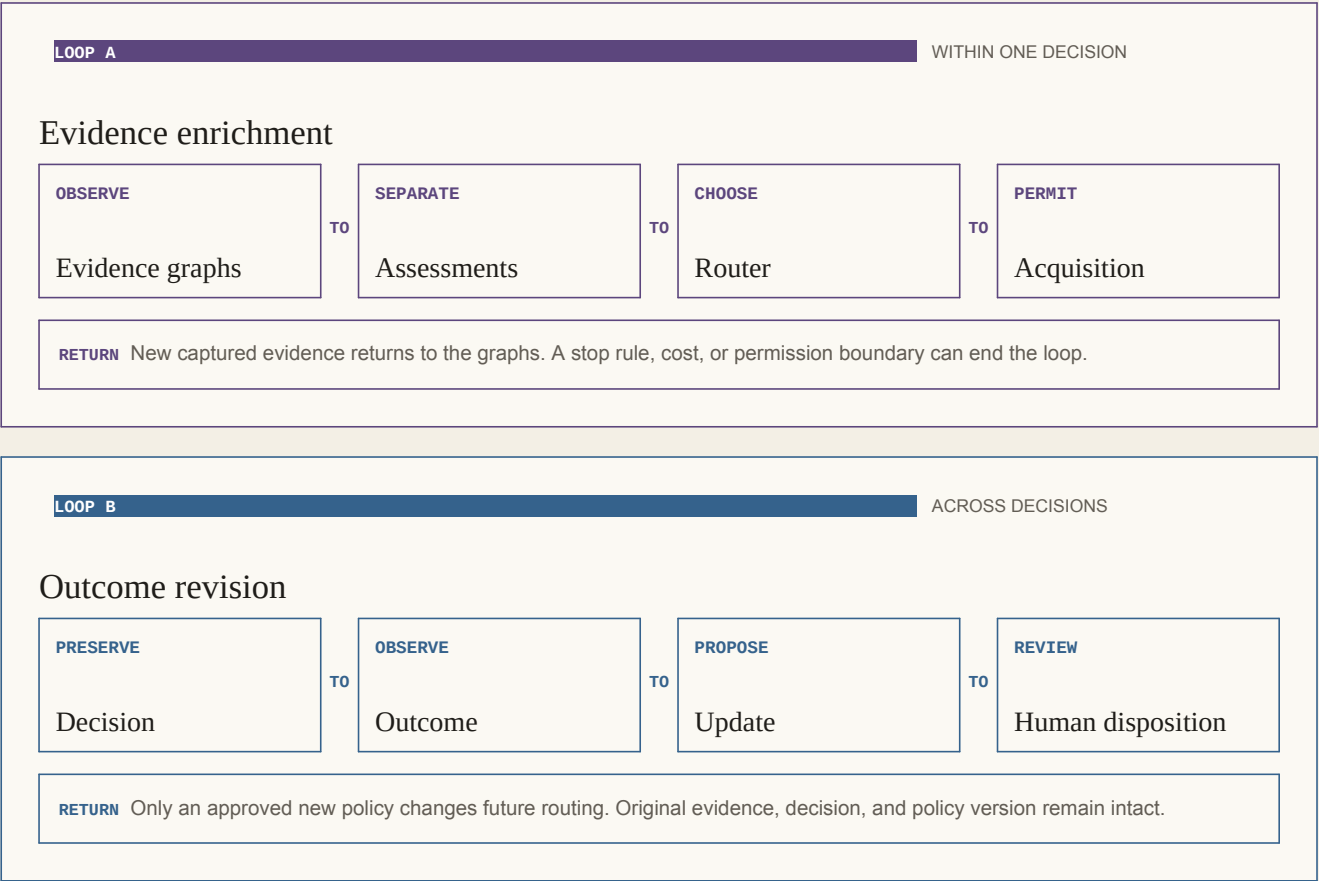
Outcome observations retain measurement source and time; policy changes are new versions linked to the evidence and approval that motivated them.

OPEN QUESTIONS

Which outcomes are observable soon enough to guide learning? How can policy improvement be separated from preference fitting?

One fast loop. One slower loop.

The framework preserves observations, interpretations, decisions, and outcomes as different record types. Revision changes a view or policy; it does not erase the evidence path that existed at decision time.



TEXT EQUIVALENT

Within a decision, a typed gap can send a cost- and permission-bounded route back to acquisition; new evidence then returns to the relationship and claim views. Across decisions, a predefined outcome can motivate a proposed policy update, but human disposition controls whether a new version affects future routing.

RECORD	EXAMPLE	REVISION RULE
OBSERVATION	A captured issue report contains a rollback account.	Preserve; append correction or supersession.
INTERPRETATION	The report may indicate a failure mode.	Revise with reason; never recast as observed fact.
DECISION	Run one synthetic-data rollback check.	Version with brief, route, owner, and cost.
OUTCOME	The predefined sandbox check passed or failed.	Record horizon and confounders; propose policy change separately.

THREE IMPLEMENTATION PATHS

Different placements, not maturity levels.

The responsibility can live in team practice, workflow coordination, model behavior, or a combination. None is inherently deeper or more defensible.

PRACTICE	SYSTEM	MODEL
Make the judgment explicit in existing work. Define the decision, separate evidential judgments, record material exclusions, and state why research stopped.	Coordinate evidence, policy, review, and memory. Assemble an inspectable context packet while preserving provenance, permissions, and correction.	Encourage information seeking and abstention. Training or agent policies may help, but cannot replace external identity, receipts, permissions, or human authority.

Nine positive articles. One launch announcement.

Illustrative example—not a reported case or result. A technical team is deciding whether to run a sandbox pilot of a data-migration tool. A flat summary sees popularity. The framework sees a chain of different questions.



MANY MENTIONS CAN PRESERVE ONE ORIGIN

In this illustration, nine observations share one known origin; two artifacts have separate roots. Repetition is neither erased nor treated as proof, and common origin does not make a report false. Illustration only; colors encode no status; not a dataset, provenance audit, or result.

A FLAT SUMMARY SAYS

“Nine positive articles make the tool look broadly validated. A pilot appears low-risk.”

THE LAYER ASKS

How many distinct origins are documented under the packet’s relation rule? Which exact claims are supported? What would change a sandbox-only decision?

STEP 1 Bound the decision Sandbox only. No production data. Ninety minutes of research.	STEP 2 Trace the material Nine positive articles paraphrase the same vendor announcement.	STEP 3 Split the claims Nine mentions remain nine observations, but they establish only one known origin under the packet's relation rule.
STEP 4 Keep judgments separate Official documentation can be authoritative and still remain vendor-linked under the stated relation rule.	STEP 5 Choose one bounded step Inspect the benchmark method and reproduce one rollback path locally.	STEP 6 Preserve the outcome A later sandbox result may change a pilot rule, not external truth.
WHAT THE PACKET SAYS The tool documents rollback. Its comparative speed claim remains vendor-supported only. Two failure reports are relevant but not prevalence evidence. Nine positive articles share one known origin. A bounded synthetic-data check is warranted before a sandbox-only pilot decision.		

A serious framework names how it could lose.

CHALLENGE 01

Old work, new label

The mechanisms reviewed here already have mature precedents. The plausible contribution is a boundary-preserving synthesis and evaluation agenda—not a new mechanism family.

CHALLENGE 02

A new gatekeeper

Any selection policy can reinforce institutional bias or erase peripheral evidence. Exclusions, unknowns, reasons, appeal, and source diversity must stay inspectable.

CHALLENGE 03

Rigor theater

Detailed provenance can trace a false claim perfectly. Lineage never upgrades correctness, independence, or permission by itself.

CHALLENGE 04

More cost than value

The architecture must beat strong simple baselines under matched time, tokens, retrieval, and review effort.

CHALLENGE 05

Decorative human review

A person placed after an opaque route may only rubber-stamp it. Review must expose the evidence path and permit consequential correction.

CHALLENGE 06

Learning the wrong lesson

Outcome feedback can encode preference, contaminated proxies, or hindsight. Updates need predefined outcomes, attribution limits, versioning, and approval.

WHAT WOULD MATERIALLY WEAKEN OR RETIRE IT FOR A NAMED TASK CLASS?

- A strong retrieval-plus-citation baseline performs equivalently at lower cost.
- Reviewers cannot reliably distinguish authority, support, independence, relevance, and action priority.
- Origin-aware grouping hides legitimate convergence as often as it prevents false corroboration.
- The interface increases overload, review time, or overreliance enough to erase any benefit.
- Outcome updates encode local preference or contaminated proxies rather than better policy.

Twelve boundaries the presentation cannot smooth away.

01 No empirical evaluation. No experiment, participant study, field outcome, or comparative performance result.	02 No claim of mechanism novelty. The components have extensive prior art, and the integration may overlap a framework not yet found.
03 No proven minimum. Eleven components are an analytical decomposition, not a required implementation count.	04 No validated constructs. Reviewers may not reliably distinguish the proposed assessment dimensions.
05 Open-world evidence remains hard. Sources change, origins are obscured, evidence is inaccessible, and support stays contested.	06 Costs and utilities are uncertain. A clean stopping rule can still stop early or encode the wrong consequence model.
07 Human control is not guaranteed. Interfaces and organizational incentives can turn review into ceremony.	08 Memory can amplify error. Retention, retrieval, and summarization can preserve stale, biased, or manipulated content.
09 The name may fail. Discrimination layer may remain ambiguous or harmful despite a technical definition.	10 Historical HTML source is bounded. The original v13 diagram is preserved and hash-verified. A rendered-DOM snapshot is archived as reference; the original standalone HTML remains unavailable.
11 Product cases are illustrative. Alpha Solver and Signal Foundry are related cases, not independent validation.	12 No publication or owner approval. This is a local review draft.

Translation, not validation

Two product settings make the responsibilities concrete. Neither product is evidence that the full framework improves outcomes or generalizes.

ALPHA SOLVER

A reasoning posture, not proof.

Boundary: repository structure and product intent do not show improved reasoning quality, safety, or outcomes.

The inspected Alpha Solver documents are intended to illustrate how a decision brief, explicit assumptions, alternatives, tool permissions, and reviewable reasoning can constrain a solution path.

SIGNAL FOUNDRY

Evidence responsibilities, not a universal model.

Boundary: those safeguards are product-specific design choices, not empirical support for eleven general responsibilities.

The inspected Signal Foundry boundary documents specify examples of immutable raw acquisition, exclusion boundaries, source-aware evidence, staged import, and separation between transcript and visual evidence.

ENTERPRISE TRANSLATION

Identity, policy, receipts, review, versioning.

The framework can map to role-based authorization, lineage systems, policy engines, approval queues, evaluation records, and risk-management controls. That mapping does not establish compliance, safety, or return on investment in any deployment.

The next artifact depends on the claim we choose to test.

The current work is a practitioner thought piece with an academic readiness path. It should not be styled as a paper until a contribution type, prior-art protocol, methods, data, participants, outcomes, and falsifiers are fixed in advance.

NARROWEST CREDIBLE FIRST PAPER · PROPOSED DESIGN, NOT A RESULT

Oracle Origin-Relation Metadata in One Frozen Model

80 development + 40 feasibility-only pilot + 300 primary + 60 locked stress bundles. Compare citation-only, rule-only, and the byte-identical rule plus stipulated relation metadata, with exact F1/F2 token parity.

Primary: all-assigned conservative-risk-coded F2–F1 false corroboration. Safety: fixed-set recall of stipulated supporting origins, with a candidate one-sided five-point non-inferiority margin. This estimates an observable condition effect—not internal reasoning, provenance discovery, human benefit, or the full layer.

01

Conceptual systems framework

Question. Can experts distinguish the typed responsibilities, and does an existing architecture already cover them?

Evidence burden. Protocol-led synthesis, construct sorting, discriminant validity, inter-rater agreement, and adversarial boundary cases.

02

Design-science artifact

Question. Can a bounded artifact make the evidence-to-action path traceable and correctable?

Evidence burden. Requirements traceability, technical invariants, usability work, a strong simple baseline, and mechanism ablations.

03

HCI / sensemaking system

Question. Which representation supports correction and calibrated reliance without overwhelming people?

Evidence burden. Representative-user studies, comprehension and correction measures, accessibility evaluation, and qualitative strategy analysis.

04

AI context / evidence architecture

Question. Does origin-aware context policy improve supported generation under matched budgets?

Evidence burden. Strong RAG and reranker baselines, known-derivation corpora, matched tokens and spend, repeated runs, ablations, and blinded adjudication.

05

Decision-support evaluation

Question. When does the framework improve decisions enough to justify its overhead?

Evidence burden. Preregistered comparative studies, validated outcomes, matched resources, blinded scoring, delayed outcomes where possible, and harm analysis.

06

Practitioner thought piece

Question. Can the framework sharpen practice without borrowing academic authority it has not earned?

Evidence burden. Transparent synthesis, disciplined status labels, concrete examples, counterarguments, and an explicit route to future evidence.

RECOMMENDED SEQUENCE

1 · construct boundary → 2 · provenance-rich benchmark → 3 · strong simple baselines → 4 · people and outcomes

Prior art narrows the claim.

The mechanisms reviewed here have mature precedents in adjacent fields. These selected primary and official sources support only the bounded points named below; they do not validate the synthesis.

[Lewis et al. \(2020\). Retrieval-Augmented Generation](#)

Retrieving external material before generation.

[Liu et al. \(2024\). Lost in the Middle](#)

Long context does not guarantee uniform use of relevant information.

[Tan et al. \(2025\). HydraRAG](#)

Integrated cross-source reasoning and corroboration make broad architecture novelty untenable.

[Ge et al. \(2025\). CONFACT](#)

Conflict and credibility information in RAG; credibility is not origin relation.

[Nematov et al. \(2025\). Source Attribution in RAG](#)

Preprint on source influence, redundancy, complementarity, and synergy; not an origin-family graph.

[Xia \(2026\). Matched Evidence Utilization](#)

Preprint precedent for observable matched-condition diagnostics rather than internal-mechanism claims.

[W3C PROV-O](#)

Entities, activities, agents, and derivation.

[Cochrane Handbook, chapter 4](#)

Collating multiple reports of one underlying study.

[Thorne et al. \(2018\). FEVER](#)

Claim and evidence-state precedent.

[Wadden et al. \(2020\). SciFact](#)

Scientific claim verification and rationales.

[Min et al. \(2023\). FActScore](#)

Atomic factual precision as evaluation precedent.

[Pirulli & Card \(1999\). Information Foraging](#)

Resource-bounded search behavior.

[Howard \(1966\). Information Value Theory](#)

Whether more information is worth its cost.

[Metzger \(2007\). Credibility assessment](#)

Contextual, multidimensional credibility assessment.

[Horvitz \(1999\). Mixed-Initiative Interfaces](#)

Uncertainty, timing, cost, and human control.

[Amershi et al. \(2019\), Human-AI Interaction](#)

Correction, control, and expectation management.

[NIST AI RMF 1.0](#)

Enterprise governance context; not product validation.

[Anthropic \(2025\), Context engineering](#)

Industry implementation context; not independent empirical support.

Interpretation guardrails

DISCRIMINATION LAYER A proposed systems responsibility for deciding what context may influence generation; not a claim about one service, model, or novel mechanism.	ATTENTION PRIORITY How urgently an item deserves inspection because of possible relevance or consequence; not truth probability.
DOMAIN SOURCE AUTHORITY Task- and claim-scoped standing to speak about a domain; not universal trust.	CLAIM SUPPORT The relationship between exact evidence and a bounded proposition; not source popularity or citation presence.
ORIGIN RELATION Whether observations share a known origin, are separately rooted as stipulated, or remain unresolved; it is not real-world epistemic independence.	RECURRENCE Repeated observation of a pattern; it does not establish distinct-origin support.
RELEVANCE Usefulness to the current brief; not general importance, correctness, or permission.	OPERATIONAL AUTHORIZATION Permission to acquire, process, disclose, retain, or act; not source competence or evidential support.
ENRICHMENT VALUE The expected benefit of another permitted operation considered with cost and risk; not action priority or acceptance.	ACTION PRIORITY The ordered permitted next step after consequence, uncertainty, cost, and authorization are considered; not a factual verdict.
ASSESSMENT ATTRIBUTES Uncertainty and possible consequence qualify judgments and routes; neither is an interchangeable master score.	PROVENANCE A trace of identity, derivation, actors, and time; not proof that content is correct.
SIGNAL CANDIDATE A derived pattern worth inspection; not a verified event or conclusion.	OWNER DISPOSITION An accountable person's recorded action or judgment; not external truth.

Does the framework make the hidden judgment easier to inspect—or merely more elaborate?

The strongest current claim is deliberately modest: this is a coherent, historically grounded synthesis worth examining. It is not complete, empirically validated, or novel as a scientific mechanism.

Current status: local owner review; not published.